



Muscle metabolism and exercise

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LEARNING OBJECTIVES

Introduction muscle metabolism & exercise

- Muscle metabolism
 - Phosphate Bond Energy
 - Anaerobic Energy Metabolism
 - Aerobic Metabolism (Mitochondria)
 - Citric Acid Cycle
- Metabolism of Glucose and Glycogen in Muscle Fibers

Introduction muscle metabolism ...

- Muscle is *body tissue* that is characterized by
 - its ability to contract, usually in response to a stimulus from the nervous system
- Of three major kinds of muscle, skeletal and cardiac muscles are high consumers of energy
- The heart is a muscular tissue
 - ✓ pump the *blood* through the circulatory system

Introduction muscle metabolism ...

- Smooth muscle is mostly found in
 - the respiratory, urinary, gastrointestinal tract, the reproductive system, and blood vessels
- Smooth muscle uses relatively little energy despite the heavy workload it has.
- Skeletal muscle is unique in energy metabolism
- In addition to its aerobic capacity, it is adapted for short-term anaerobic activity

Introduction muscle metabolism ...

- **Chemical energy** is stored in muscle cells as ATP and creatine phosphate
- ATP provides the energy for all forms of muscle work
- ATPases, the enzymes that break down ;
 - ATP and release the energy for muscle work and metabolism

Introduction muscle metabolism ...

- In skeletal muscle, contractions are normally associated with depolarization of the plasma membrane,
- which initiates the release of calcium ions from intracellular stores of the sarcoplasmic reticulum
- Calcium ions bind to troponin C, a regulatory protein associated with the thin filaments,
 - producing a change in protein conformation
- The contraction ceases when calcium ions are taken up by the sarcoplasmic reticulum through the operation of an ATP-driven pump, commonly known as Ca^{2+} ATPase

Muscle metabolism...

1. Phosphate Bond Energy

- Energy in **skeletal muscle** is derived mostly from glucose and fatty acids
- It is also **stored** in significant amounts as glycogen and triglycerides, respectively, in the muscle fibers
- The chemical energy **trapped within the bonds** of
 - the carbohydrate, lipid, and protein molecules
- & is extracted as adenosine triphosphate (ATP), an immediate source of energy
 - ATP consists of an adenosine molecule linked to sugar & three phosphates

Muscle metabolism...

- The bonds that link the molecule to the outer most phosphates are called high-energy phosphate bonds, since their hydrolysis (when ATP h/l with water) releases 7.3kcal of energy
- This reaction is catalyzed by an enzyme called adenosine triphosphatase (ATPase), and the product is an adenosine molecule containing
 - two phosphate groups called adenosine diphosphate (ADP).

Muscle metabolism...

- Additional energy is produced when the second phosphate bond is hydrolyzed and
 - a single phosphate containing adenosine monophosphate (AMP) is the end product.
 - $\text{ATP} + \text{H}_2\text{O} \xrightarrow{\text{ATPase}} \text{ADP} + \text{P} + 7.3 \text{ kcal. mol}^{-1}$
- The stores of ATP itself are sufficient to provide energy for a few seconds
- The total ATP store in human skeletal muscle is approx 80 g.

Muscle metabolism...

- ATP is resynthesized at the rate of its consumption, through three main mechanisms:
 - short-term high-energy phosphates (creatine phosphate),
 - medium-term supply (via anaerobic glycolysis), and
 - long-term supply (via the oxidative phosphorylation of glucose and fatty acids to water and CO₂)
- Major sources of ATP resynthesis are the oxidation of lipids and carbohydrates, which are slow and constant processes

Muscle metabolism...

- A rapid turnover of ATP is maintained without the use of oxygen through
 - CP, a high-energy phosphate compound.
- Energy is released with the breakdown of CP, for
 - the immediate synthesis of ATP
- ATP and CP together (called a phosphagen system) are;
 - a critically important source of muscle contraction, especially in athletic activities,
 - that need high power for a short period of time,
 - such as the quick starts of sprinters and high jumpers

Muscle metabolism...

- ATP and CP concentrations are therefore fairly constant (approximately 5 mmol L⁻¹ and 30 mmol L⁻¹ respectively)
- In the recovery phase after muscle contractions,
 - CP is resynthesized from its breakdown products creatine and inorganic phosphate at the cost of ATP
- The energy required for the phosphagen replenishment comes from the aerobic metabolism

Muscle metabolism...

2. Anaerobic Energy Metabolism

- Glycolysis is the pathway for the catabolism of glucose, occurring in the cytosol.
- Glycolysis is unique in that
 - it can use oxygen if available (pyruvate $\rightarrow$ acetyl-CoA), or
 - function without oxygen (pyruvate $\rightarrow$ lactate)
- The relative role of glycolysis as a source of energy varies between tissues
 - for example, slight in the heart, and major in the brain and red cells

Muscle metabolism...

- In skeletal muscle, glycolysis permits high performance,
 - when aerobic metabolism alone is not sufficient
- In skeletal muscle at rest, glycolysis provides
 - the acetyl-CoA used in the citric acid cycle
- In the process, 6 carbon glucose is catabolized to 3 carbon pyruvate and then to acetyl-CoA,
- NADH formed through glycolysis is transported via
 - the malate shuttle into the mitochondria and oxidized in the respiratory chain, with a net yield of 3 ATP per NADH

Muscle metabolism...

- in the complete oxidation of 1 mole of glucose under aerobic conditions,
 - glycolysis yields 8 ATP and the citric acid cycle 30 ATP
- Skeletal muscle is also readily subjected to anaerobiosis
- This property allows for short term performance far exceeding the levels that can be handled aerobically
- **Two of the three** mechanisms of ATP re-synthesis are through
 - anaerobic metabolism, which means without the use of oxygen

Muscle metabolism...

- Anaerobic energy metabolism, also called anaerobic glycolysis, involves
 - the incomplete breakdown of glucose to lactic acid pathways
- Anaerobic glycolysis is involved during muscular activities that last for as short a period or a few minutes, with high energy demand
 - Where aerobic metabolism is not adequate for the energy requirements
- This process takes place in the cytoplasm,
 - Although there is rapid ATP synthesis
 - anaerobic glycolysis is less efficient than aerobic glycolysis.

Muscle metabolism...

- An accumulation of lactic acid decreases intracellular pH, which
 - inhibits the activity of phosphofructokinase, rate-limiting enzyme of glycolysis
- Furthermore muscle NADH decreases during low-intensity exercise, but
- it increases , as -intensity exercise high
- The increase in muscle NADH can result from
 - limited availability of O₂ in the **contracting muscle**

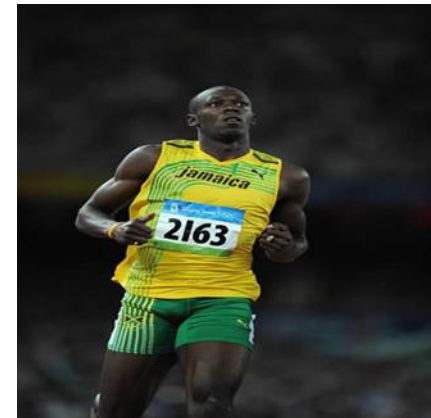
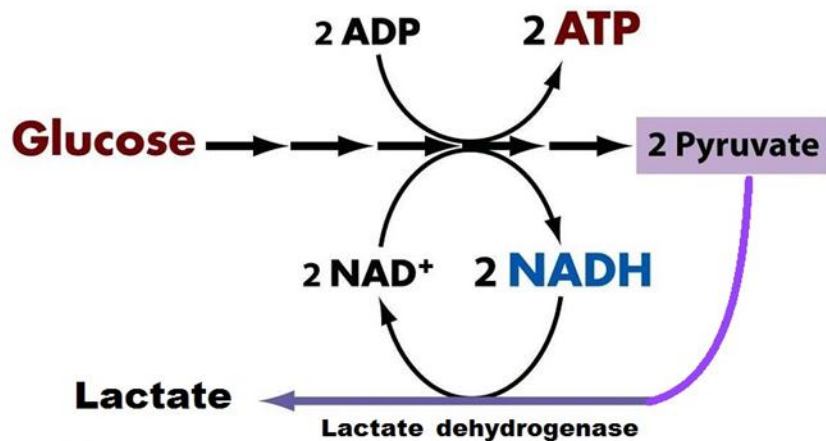
Muscle metabolism...

- During strenuous exercise, Oxygen in the Skeletal muscles is depleted
- Lactate released to the blood may be taken up by the liver & converted to back to glucose via gluconeogenesis,
- After exercise, a person breathes heavily to re-pay the oxygen debt and reform pyruvate in the liver
- Or may be taken up skeletal muscle and converted back to pyruvate, via Lactate Dehydrogenase
 - after exercise, which may be oxidized in Citric Acid Cycle

Muscle metabolism...

- The cornea, lens, and regions of the retina have a limited blood supply and also lack mitochondria and depend on glycolysis as the major mechanism for ATP production
- Tissues dependent primarily on anaerobic glycolysis for ATP production

Formation of lactate allows NAD^+ to be reformed and reused for glycolysis



Muscle metabolism...

- Oxidized NAD^+ can act as a hydrogen acceptor, allowing glycolysis to proceed
- ATP formation by anaerobic process is, however, at a high cost
- Oxidation of 1 mole of glucose results in a net yield of only 2 moles of ATP
- Increased lactic acid production can hamper
 - the function of muscle fibers themselves &
 - cells of the connective tissue

Differences between aerobic and anaerobic glycolysis:

	Aerobic glycolysis	Anaerobic
1. End product	Pyruvate	Lactate
2 .Energy	8 ATP	2 ATP
3.Regenerationof NAD⁺	Through respiration chain (ETC) in mitochondria	Through Lactate formation
4. Availability of Pyruvate to TCA in mitochondria	2 Pyruvate Available & can oxidize to give ATP	Not available to TCA as lactate is cytoplasmic substrate

Muscle metabolism...

3. Mitochondria and Aerobic Metabolism

- Oxygen is transported by the respiratory chain to the site of oxidation in active tissues
- Oxygen is transported in the circulation bound to hemoglobin of erythrocytes, while substrates are transported in the plasma
- The supply of oxygen must be continuous, because there are only minimal oxygen stores in the body of most mammalian species, while
 - substrates are stored in significant quantities both within muscle cells and in tissue substrate stores

Muscle metabolism...

3.1. Mitochondrial Oxidative Phosphorylation

- Aerobic energy metabolism takes place in the mitochondria, and it results in the greatest release of energy
- As the name implies, though, it requires oxygen.
- Aerobic glycolysis is the most efficient method of energy production
- Anaerobically only 2 moles of ATP are synthesized from 1 mole of glucose , whereas
 - a total of 38 moles of ATP could be synthesized from glucose in the presence of oxygen
 - The net yield is 36 moles of ATP when 2 moles of ATP consumption is subtracted of 38 molecules

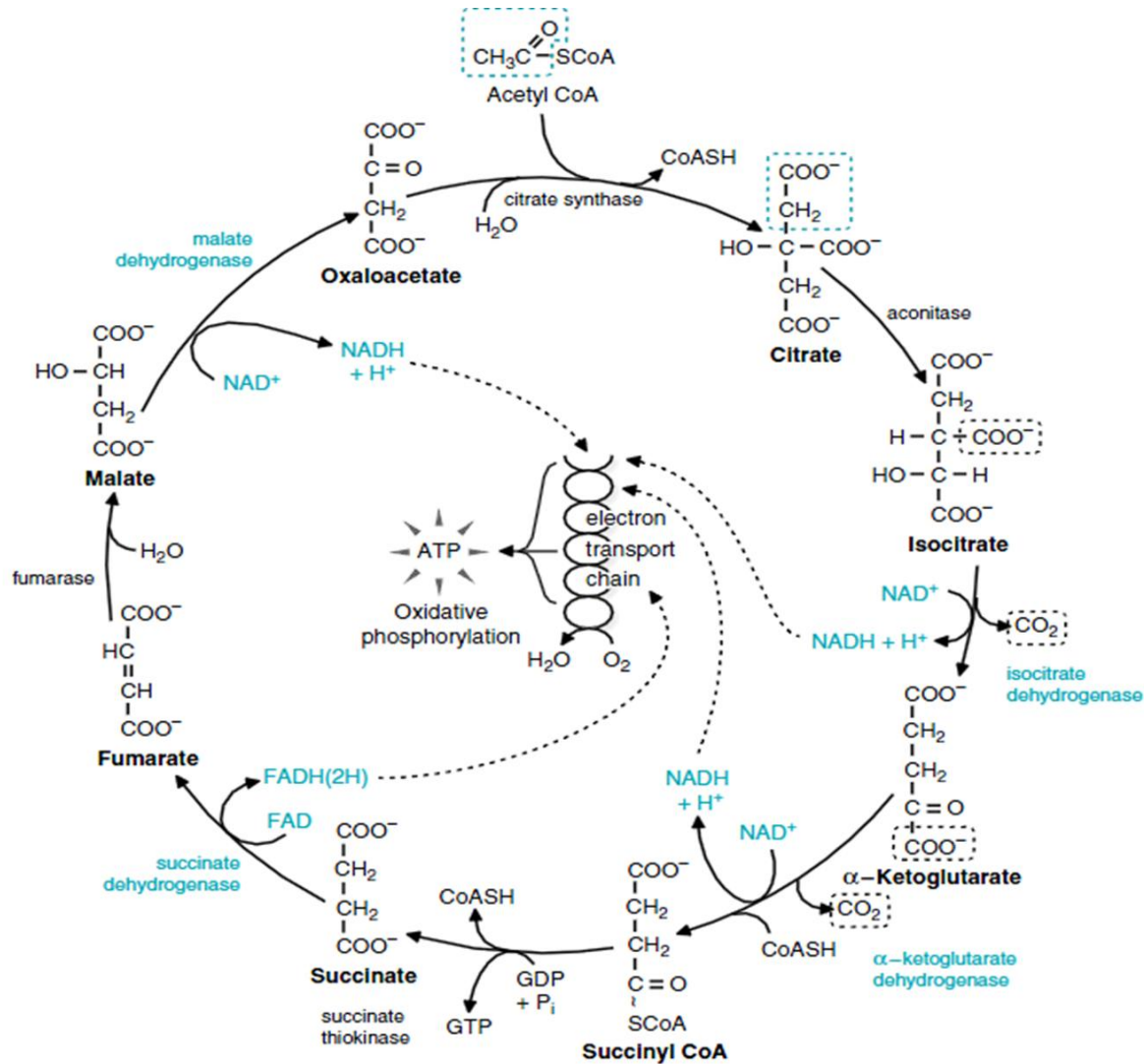
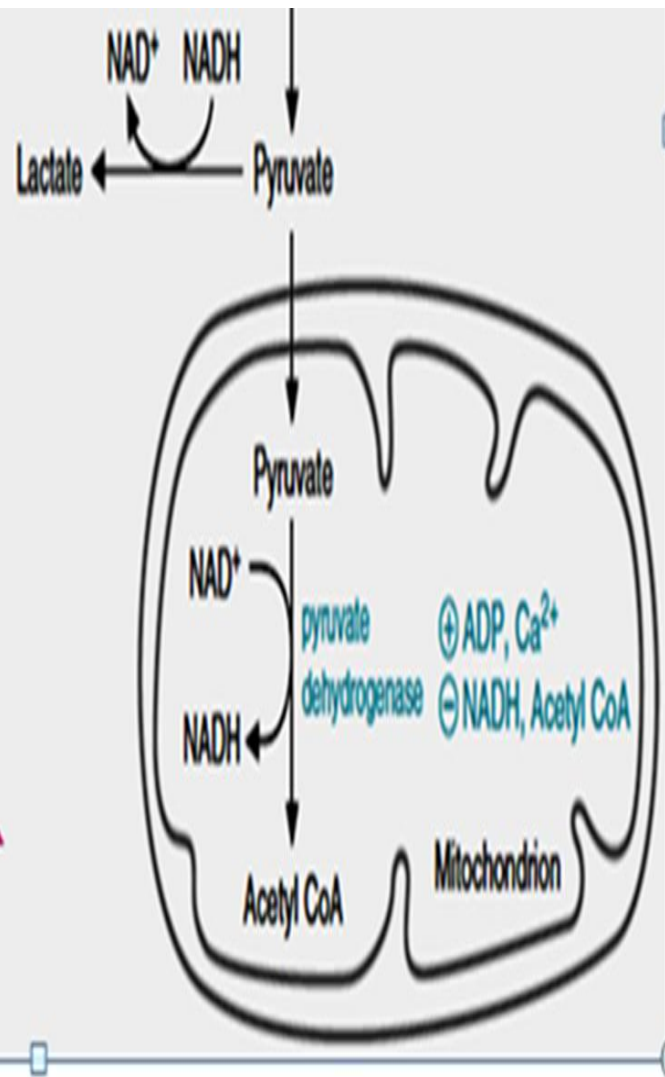
Muscle metabolism...

- In mitochondria of muscle cells, hydrogen atoms are extracted from the acetyl-coA that are fed to citric acid cycle,
 - By a process called oxidative phosphorylation.

Muscle metabolism...

3.2. Citric Acid Cycle

- The citric acid cycle is the common final pathway for glucose (glycolysis), fatty acid (beta-oxidation), and protein catabolism
- The citric acid cycle is a series of reactions in the mitochondria, which bring about the catabolism of acetyl-CoA,
 - liberating hydrogen equivalents that are used in oxidative phosphorylation for the generation of ATP from ADP.
-



In the citric acid cycle, acetyl-CoA condenses with oxaloacetate to form citrate.
reduced electron carriers (NADH , and FADH_2)

Muscle metabolism...

- Both at rest and during exercise, fatty acid beta-oxidation and glycolysis provide over 95% of acetyl-CoA entering the citric acid cycle.
- one turn of the citric acid cycle generates 12 ATP, 3 NADH, 1 FADH₂, 1 GTP, and 2 CO₂ are produced
- Purpose of NADH + H⁺ & FADH₂ molecules is to take energy rich electrons and protons from substrates of Krebs cycle to ETCS
- What will happen when energy rich electrons & protons are passing through ETCS?
 - Oxygen will combine the proton, & H₂O is produced
- The reducing equivalents subsequently undergo oxidative phosphorylation

Muscle metabolism...

4. Metabolism of Glucose and Glycogen in Muscle Fibers

- **Skeletal muscle** derives glucose from
 - glycogenolysis or
 - from the blood
- Glucose can be stored as glycogen up to the level of 4–5% of the wet weight of the muscle tissue
- Glycogen is the major supply of glucose during moderate and intensive exercise
- Muscle extracts glucose from blood through insulin-dependent mechanisms

Muscle metabolism...

- Exercise increases skeletal muscle insulin sensitivity
- Muscle glucose transporter levels such as GLUT4, an important limiting factor of glucose utilization, and glycogen synthase activity, increase in response to exercise training
- The GLUT 4 is the major facilitative glucose transporter in the skeletal muscles, adipose tissues and other peripheral tissues

THANK YOU !